# ML Report:
# VoiceBridge

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## 1 Introduction

The VoiceBridge project addresses the digital accessibility gap faced by individuals with speech impairments resulting from neurological conditions like Parkinson's disease, stroke, or cerebral palsy. While modern Automatic Speech Recognition (ASR) models have advanced significantly, flagship architectures such as Whisper and wav2vec often exhibit drastically reduced performance when processing atypical or dysarthric speech. This creates significant barriers to independence, preventing users from performing everyday digital tasks like messaging or browsing the internet. The goal of this initiative is to bridge this gap by fine-tuning a machine learning model to reliably interpret impaired speech and map it to actionable digital commands.

The system's objective is to capture impaired audio through a standard microphone and process it via a custom-trained machine learning model. The resulting text output is not merely for transcription but is designed to be mapped into actionable commands. By transforming these inputs into device-level controls—such as sending messages or operating applications—VoiceBridge aims to restore autonomy to users in everyday environments, including homes and clinics. This objective heightens the need for an accurate model to provide transcripts, as commands are often only a couple words in length. Due to the lack of context available to the model when processing short phrases, high recognition accuracy and low word error rate (WER) are vital for the systems relying on these model's performance.

## 2 Related Work

Gemini said A semi-systematic review of ASR approaches for impaired speech indicates that fine-tuning large, pre-trained foundation models consistently outperforms models trained solely on typical speech (SpeechBuddies, 2026). Research suggests that while full fine-tuning of all model layers can reduce WER by 30–60%, it is computationally expensive and prone to overfitting when data is scarce (SpeechBuddies, 2026). Consequently, parameter-efficient adaptation methods, such as Low-Rank Adaptation (LoRA) and adapters, have emerged as practical alternatives. As noted by Pokel et al. (2025), LoRA provides a reported WER reduction of 7–40

Beyond model architecture, the literature emphasizes that preprocessing and feature engineering are critical for handling the irregular articulation and non-stationary noise often found in impaired speech recordings (SpeechBuddies, 2026). Essential techniques include:

- **Temporal Normalization:** Methods such as Dynamic Time Warping (DTW) and Temporal Structure Normalization (TSN) address timing irregularities in dysarthric speech (nor, 2025).
- **Denoising:** DFT-domain subspace decomposition helps separate speech from background noise in uncontrolled environments (ScienceDirect, 2024).
- **Phoneme-aware Modeling:** Extracting phonetic representations (e.g., via Wav2Vec2-Phoneme) improves the precision of short command-based interactions (Hugging Face).

## 3 Dataset

You should write about your dataset here, following the guidelines regarding item 1. This section may be 0.5-1 pages. Depending on your specific dataset, you may want to include subsections for the preprocessing, annotation, etc.

## 4 Features

Describe any features you used for your model, or how your data was input to your model. Are you doing feature engineering or feature selection? Are you learning embeddings? Is it all part of one neural network? Refer to item 3. This may range from 0.25 pages to 0.5 pages.

## 5 Implementation

The VoiceBridge implementation focuses on a parameter-efficient fine-tuning pipeline using Whisper-small as the foundation.

### 5.1 Data Preparation and Preprocessing

The system processes audio samples standardized to a sampling rate of 16,000 Hz. A custom `AudioTextDataset` class manages the data pipeline:

- **Audio Standardization:** Stereo signals are converted to mono by averaging channels to ensure consistency across different recording hardware.

- **Feature Extraction:** The `WhisperProcessor` extracts spectral features from the waveforms, while the associated tokenizer processes transcripts into input IDs.

- **Batching:** A `DataCollatorSpeechSeq2SeqWithPadding` handles dynamic padding for both input features and labels. Padding in labels is masked with a value of -100 to ensure it does not contribute to the loss calculation during training.

### 5.2 Fine-Tuning with Low-Rank Adaptation (LoRA)

To adapt the model to atypical speech patterns without retraining all 244 million parameters, the project utilizes the `peft` library to implement LoRA.

- **Target Modules:** Adapters are applied specifically to the Query (`q_proj`) and Value (`v_proj`) projection matrices within the transformer's attention mechanism (Pokel et al., 2025).

- **Hyperparameters:**
  - **Rank ($r$):** Set to 16, providing a balance between model capacity and efficiency.
  - **Alpha ($\alpha$):** Set to 32 as a scaling factor for the low-rank updates.
  - **Dropout:** A rate of 0.1 is applied to the LoRA layers to prevent overfitting on the relatively small impaired speech datasets.

- **Architecture Patching:** A `patched_forward` method is used to wrap the base model, ensuring that the `peft` model correctly handles `input_features` during the training pass.

### 5.3 Training Configuration

Training is conducted over 3 epochs using the `AdamW` optimizer.

- **Learning Rate:** A stable rate of $1 \times 10^{-4}$ is maintained.

- **Checkpointing:** The system saves the model state dict to a `./checkpoints` directory every 100 batches, facilitating recovery and performance tracking across the training lifecycle.

- **Batch Size:** A batch size of 4 is utilized to fit within the memory constraints of standard GPU environments.

## 6 Results and Evaluation

How are you evaluating your model? What results do you have so far? What are your baselines? Refer to item 5. This may take around 0.5 pages.

## 7 Feedback and Plans

The VoiceBridge project roadmap is strategically bifurcated into a **Minimum Viable Product (MVP)** and a set of **Stretch Goals** to ensure technical feasibility within the Capstone timeline while maintaining a path toward a comprehensive accessibility solution.

### 7.1 Core Performance Benchmarks

Success for the initial deployment is defined by two primary technical metrics:

- **Transcription Accuracy:** A target of **70%+ accuracy** on dysarthric speech. This represents a significant improvement over baseline Whisper-small performance for impaired vocalizations.

- **Real-Time Latency:** Processing delay must remain **under 1 second**. This threshold is critical for interactive digital tasks, ensuring that the feedback loop between speech and device action feels natural and responsive for the user.

### 7.2 User Experience and Feedback Loops

To build user trust and mitigate the impact of ASR errors, the system incorporates a **non-disruptive feedback mechanism**. Transcribed text is displayed in a dedicated interface, allowing for immediate user verification. If a command is misinterpreted, the system provides clear pathways for manual correction, which not only assists the user but also provides potential data points for future iterative model refinement.

### 7.3 Future Personalization and Environmental Robustness

Expanding beyond the MVP, the team has identified several high-impact stretch goals:

- **Custom Command Mapping:** Users will be able to define personalized "vocal shortcuts." For instance, a user with specific slurred phonemes can map a consistent idiosyncratic phrase to a high-level digital trigger, such as “Open Browser” or “Emergency Alert”

- **Acoustic Robustness:** Current research focuses on enhancing “Robustness to Noise” (ScienceDirect, 2024). This ensures the model remains functional in non-idealized environments, such as busy public clinics or domestic settings with ambient background noise.

### 7.4 Project Timeline and Ethical Data Lifecycle

The project is currently in the data collection and architectural refinement phase.

- **Phase 1 (Summer 2026):** Participant recruitment and audio data collection for fine-tuning will continue, coordinated with high-speed network support via ORION for large-scale data transfer.

- **Phase 2 (Fall 2026):** Detailed architectural reviews with technical advisors, including Pavel, will guide the final system integration and performance audit.

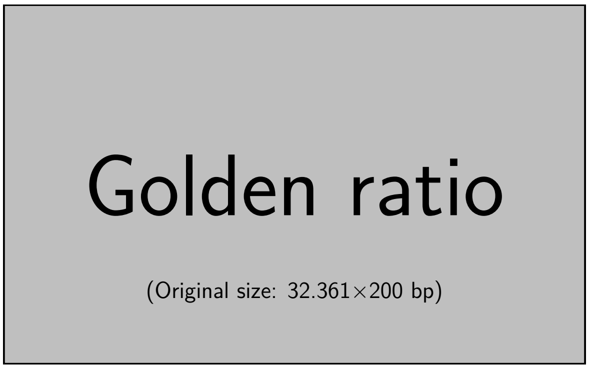


Figure 1: A figure with a caption that runs for more than one line. Example image is usually available through the `mwe` package without even mentioning it in the preamble.

- **Phase 3 (Winter 2027):** The project is slated for official completion. In accordance with established ethics protocols, all raw research data will be permanently purged to protect participant privacy and comply with data retention agreements.

### 7.5 Tables and figures

See Table **??** for an example of a table and its caption. See Figure 1 for an example of a figure and its caption.

## 8 References

### Team Contributions

Write in this section a few sentences describing the contributions of each team member. What did each member work on? Refer to item 7.

### References

2025. Normalization through fine-tuning: understanding wav2vec 2.0 embeddings for phonetic analysis. *arXiv 2503.04814*.

Hugging Face. *Wav2Vec2-Phoneme Documentation*.

N. Pokel, P. Moure, R. Boehringer, S.-C. Liu, and Y. Gao. 2025. Variational low-rank adaptation for personalized impaired speech recognition. *arXiv preprint arXiv:2509.20397*.

ScienceDirect. 2024. Dysarthric speech preprocessing under low-resource/noisy conditions. *Speech Communication*.

SpeechBuddies. 2026. Literature review - voicebridge: Semi-systematic review of asr approaches for impaired speech. Technical report, McMaster University.

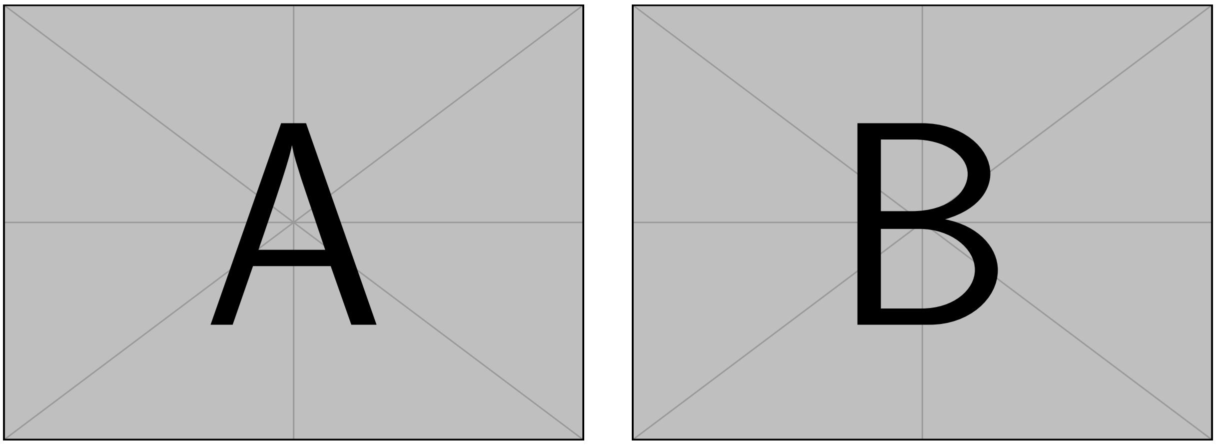


Figure 2: A minimal working example to demonstrate how to place two images side-by-side.